

MICROBIOLOGY – SHORT EXAM NOTES

Basic Equipment in Microbiology • Demonstration of Compound Microscope (Ananthanarayan & Paniker)

1. BASIC EQUIPMENT IN MICROBIOLOGY

1. Inoculating Loop and Straight Wire

Use: to transfer and inoculate specimens or bacterial growth onto culture plates and tubes (streak, stab and stroke cultures). **Material:** **platinum or nichrome wire (24 SWG)** fixed in a metal or glass handle (Kolle's holder). The **loop** (about 2–4 mm diameter) is used for streaking plates and broth; the **straight wire** is used for stab cultures and for picking single colonies. **Sterilisation: flaming** – held in a Bunsen flame until **red hot**, then cooled before use. A loop carrying infective material should be dipped in disinfectant (or heated gradually from the shank) before flaming to prevent **spattering**. Disposable sterile plastic loops are also available.

2. Bunsen Burner and Spirit Lamp

Bunsen burner – a gas burner used for **flaming** (sterilisation by dry heat). Loops, wires, tips of forceps and searing spatulas are sterilised by holding them in the flame till **red hot**. It also creates an updraft that reduces airborne contamination around the working area; mouths of tubes and flasks are flamed while inoculating. The hottest part is the tip of the inner blue cone. **Spirit lamp** – a glass lamp with a wick burning **alcohol (methylated spirit)**; used where gas supply is not available, for flaming loops, needles and slide fixation. Its flame is less hot than a Bunsen flame.

Book reference: flaming is described under dry heat in Ch. 3; spirit lamp details are standard practical content.

3. Hot-Air Oven (HAO)

Principle: sterilisation by **dry heat** through **conduction**; kills by **oxidation** of cell constituents and protein denaturation. It is the most widely used method of dry-heat sterilisation. **Holding time: 160°C for 2 hours** (also 170°C/1 h or 180°C/30 min). **Construction:** electrically heated chamber with a **fan** for even distribution of air and a thermostat.

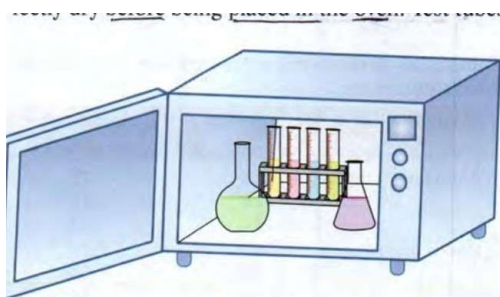


Fig. 3.1 Hot-air oven (from the book)

- **Used for:** glassware (Petri dishes, flasks, pipettes, test tubes), all-glass syringes, forceps, scissors, scalpels, **liquid paraffin, dusting powder, fats and grease**.
- **Precautions:** do not overload; allow free air circulation; glassware must be **perfectly dry**; wrap tubes/flasks in craft paper; rubber (except silicone) cannot be used; cotton plugs char at 180°C; allow to **cool slowly (~2 h)** before opening, or glassware may crack.
- **Sterilisation control:** physical – thermocouples; chemical – **Browne's tube (green spot)** turns green after 60 min at 160°C; biological – spores of non-toxicogenic *Clostridium tetani* or *Bacillus subtilis* subsp. *niger*.

4. Autoclave

Principle: steam under pressure (moist heat). When pressure rises in a closed vessel, water boils at a higher temperature. Steam condensing on a cooler surface releases **latent heat**, and the condensed water provides moist heat that kills by **denaturation and coagulation of proteins**. Standard cycle: **121°C at 15 lb/in² (1 atm above normal) for 15 minutes** (also 126°C/10 min, 134°C/3 min). Kills vegetative forms and spores.

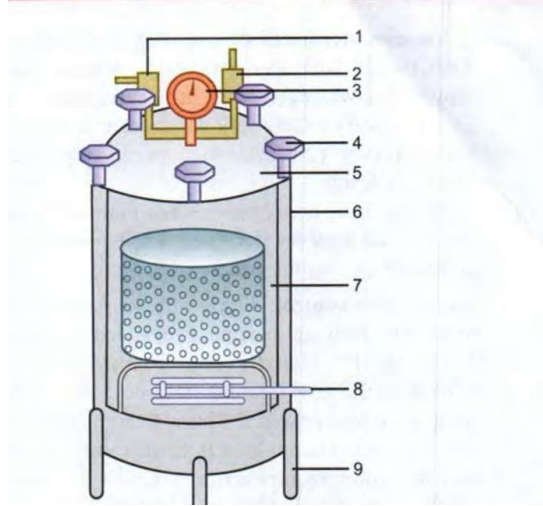


Fig. 3.3 A simple autoclave: 1. release valve 2. safety valve 3. pressure gauge 4. tightening screws 5. cover 6. body 7. bin containing material 8. heating element 9. stand (from the book)

- **Construction:** vertical/horizontal cylinder of gunmetal or stainless steel; lid fastened by screw clamps; **discharge (air) tap, pressure gauge, safety valve**; electric heating.
- **Procedure:** add water, load material, close lid with discharge tap open, heat until **all air is expelled** (no bubbles when steam is led into water), close tap, let pressure rise; start the **holding time** when the set pressure is reached; switch off and let it cool until pressure falls to atmospheric before opening the tap slowly.
- **Types:** gravity displacement type and **high-vacuum** sterilisers (faster, dry loads). A **pressure cooker** acts as a miniature autoclave.
- **Used for:** culture media, dressings, linen, instruments, laboratory ware, gloves, and **decontamination of infectious waste**.
- **Sterilisation control:** spores of *Geobacillus stearothermophilus* (killed at 121°C in 12 min) – biological indicator; also autoclave tape and chemical indicators.

5. Filters – Candle Filters

Filtration removes bacteria from **heat-labile liquids** such as sera, sugar and antibiotic solutions; it makes the fluid bacteria-free but **not virus-free**. **Candle filters** are hollow candle-shaped filters of different porosity, used for water purification. Two types: **unglazed ceramic** (**Chamberland**, Doulton) and **diatomaceous earth** (**Berkefeld**, Mandler). Other filters are **asbestos** (**Seitz**) – no longer used as carcinogenic; **sintered glass** – brittle and expensive; and **membrane filters** (cellulose esters, **0.22 µm** pore size) – now most widely used for sterilisation and sterility testing.

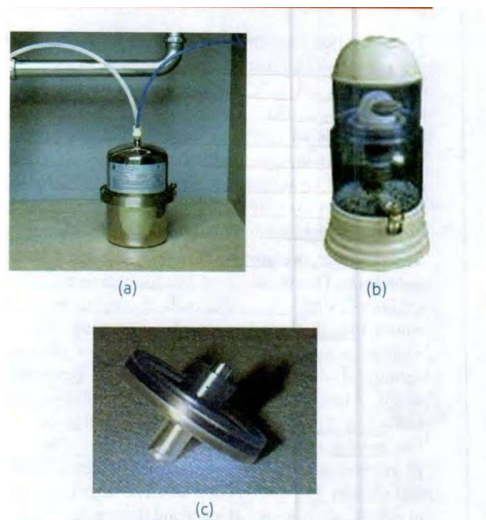


Fig. 3.4 (a) Asbestos filter (b) Sintered glass filter (c) Membrane filter (from the book)

6. Incubator

An insulated chamber that maintains a **constant temperature** (usually **37°C** for bacteria) for the growth of cultures. Inoculated plates and tubes are incubated **overnight or longer** before reading colonies or turbidity. It has a **thermostat**, thermometer, heating element, double walls with insulation and perforated shelves; plates are kept **inverted** (lid down) to prevent condensation dropping on the medium. Variants include the **CO₂ incubator** (5–10% CO₂ for capnophilic organisms) and BOD incubators (lower temperatures, e.g. fungi at 25–30°C). It is cleaned with **methyl alcohol**, which is effective against fungal spores.

Book mentions incubation at 37°C in a bacteriological incubator and cleaning with methyl alcohol; construction details are standard practical content.

7. Centrifuge

Principle: separates particles by **centrifugal force** – heavier particles sediment at the bottom as the rotor spins. Speed is expressed as **rpm** or **relative centrifugal force (g)**. **Uses:** concentration of specimens (e.g. sputum digested by **Petroff's method** is **centrifuged at 3000 rpm for 20 min**), urine deposit for microscopy, separation of **serum/plasma** from blood, and sedimentation of cells. **Precautions:** tubes must be **balanced** (equal weight placed opposite), lid closed before starting, never open while rotating; use sealed buckets for infectious material to avoid aerosols.

Book mentions centrifugation in specimen processing; the equipment description is standard practical content.

8. Laminar Air Flow (LAF) and Biosafety Cabinet (BSC)

Laminar air flow cabinet – air is passed through a **HEPA filter** (removes particles $\geq 0.3 \mu\text{m}$) and flows in parallel, uniform streams over the work area, keeping it **sterile**. It **protects the product/culture only**, not the worker, so it is used for **media pouring, plate preparation and tissue culture** – not for handling pathogens. It may be horizontal or vertical. Laminar flow ventilation is also used in operation theatres (up to **300 air changes per hour**, $\leq 10 \text{ cfu/m}^3$).

Biosafety cabinet (BSC) – a ventilated enclosure with **inward airflow and HEPA-filtered exhaust**, which protects the **worker, environment and (Class II/III) the sample**. It is used for handling infectious material; UV light is used to disinfect the cabinet between uses.

Class	Features	Protects
Class I	Open front; inward air; HEPA-filtered exhaust	Worker & environment (not sample)

Class	Features	Protects
Class II	Open front; HEPA-filtered downflow + exhaust; most common in clinical labs (e.g. TB processing)	Worker, environment and sample
Class III	Totally enclosed, gas-tight glove box; for BSL-4 agents	Maximum protection

Laminar air flow	Biosafety cabinet
Protects the product only	Protects worker, environment ($\pm$ product)
Air blows towards the worker	Air drawn inward, away from worker
HEPA-filtered supply air	HEPA-filtered supply and exhaust
Media preparation, sterile work	Handling pathogens (TB, anthrax, fungi)

Book mentions Class II BSC for TB specimens and BSL II/III cabinets for anthrax; classification details are standard practical content.

★ EXAM FOCUS

- Short notes: **Autoclave** (principle, parts, procedure, control), **Hot-air oven**, **Filtration / candle filters**, **Biosafety cabinet vs laminar flow**.
- One-liners: autoclave – **121°C, 15 lb, 15 min**, indicator *G. stearothermophilus*; HAO – **160°C, 2 h**, Browne's tube / *B. subtilis* var. *niger*; membrane filter – **0.22 μm** ; loop – platinum/nichrome, 24 SWG.

2. DEMONSTRATION OF COMPOUND MICROSCOPE

The **compound (bright-field) light microscope** uses two lens systems – the **objective** and the **ocular (eyepiece)** – to produce a magnified image of a specimen illuminated by transmitted light. **Total magnification = objective $\times$ eyepiece** (e.g. $100\times \times 10\times = 1000\times$). Its **resolving power** is limited by the wavelength of light; the limit of resolution is about **0.2–0.3 μm (300 nm)**.

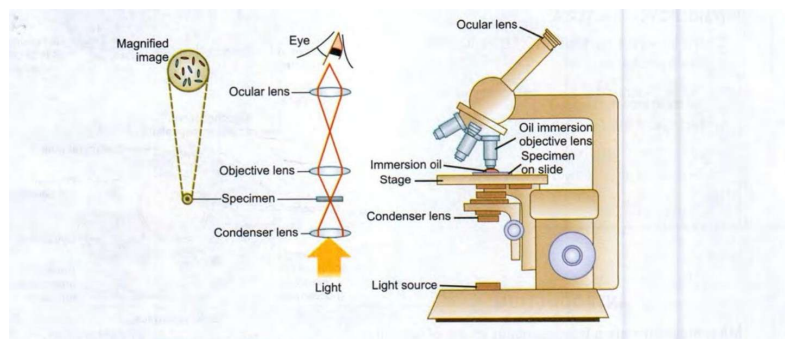


Fig. 2.2 (a) Principle of bright-field (light) microscopy (from the book)

Parts of the Compound Microscope

Part	Function
Base & arm (limb)	Support; the arm is used to carry the microscope
Body tube / head	Holds the eyepiece; monocular or binocular
Eyepiece (ocular)	Magnifies the image formed by the objective (usually $10\times$)
Revolving nosepiece	Holds the objectives and rotates to change them

Part	Function
Objectives	Low power 10×, high power 40× (high dry), oil immersion 100×
Stage with clips / mechanical stage	Holds and moves the slide
Condenser	Focuses light on the specimen; raised fully for oil immersion
Iris diaphragm	Controls the amount of light / contrast
Mirror / light source	Illumination (concave mirror or built-in lamp)
Coarse & fine adjustment knobs	Coarse for rough focus; fine for sharp focus

Oil Immersion Objective

Used for examining **stained bacterial smears**. A drop of **cedarwood oil / immersion oil** is placed between the slide and the 100× objective. The oil has nearly the **same refractive index as glass (1.515)**, so light rays passing from the slide are not refracted away; more light enters the objective, increasing **numerical aperture and resolution**. Without oil, light is lost by refraction in air.

Procedure (Demonstration)

1. Place the microscope on a flat table; switch on the light / adjust the mirror.
2. Start with the **low-power (10×)** objective; raise the condenser and open the diaphragm.
3. Place the slide on the stage and secure it with clips.
4. Focus with the **coarse adjustment**, then sharpen with **fine adjustment**; locate the area of interest.
5. For bacteria: put a drop of **immersion oil** on the smear, rotate the **100× oil immersion** objective into place, lower it into the oil while watching from the side, then focus with the **fine adjustment only**.
6. After use, lower the stage, remove the slide, **wipe oil from the objective with lens paper** (xylene if needed), set to low power and cover the microscope.

Care and Precautions

- Carry holding the **arm and base** with both hands.
- Never use the coarse adjustment with the high-power or oil immersion objective; avoid touching lenses with fingers.
- Clean lenses only with lens paper; always remove oil after use; store covered in a dry place.

★ EXAM FOCUS

- Viva: **resolving power ~0.2 μm**; total magnification = objective × eyepiece; **why oil is used**; functions of condenser and diaphragm; objective used for bacteria – **100× oil immersion**.

The book describes the light microscope and its principle (Ch. 2, Fig. 2.2a). The parts table, demonstration steps and care of the microscope are standard practical content, as the textbook does not detail them.